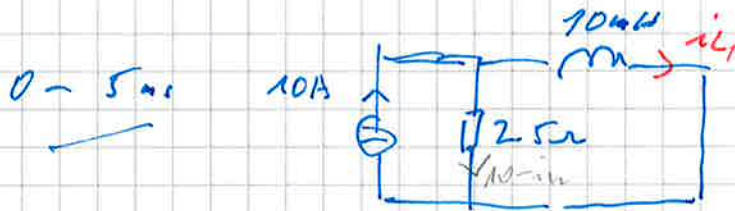


hsp Februar 201, ach1



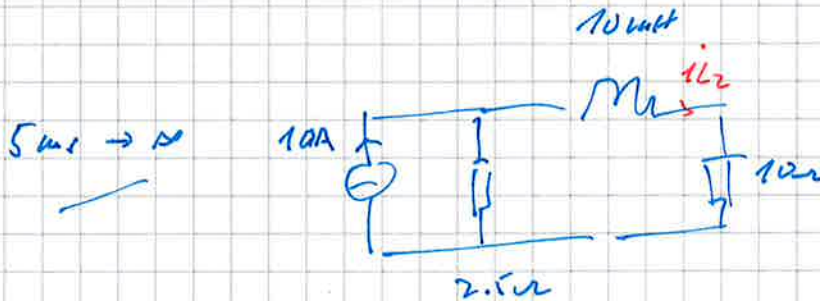
$$\tau_1 = 4 \text{ ms}$$

$$i_L(0) = 0$$

$$i_L(\infty) = 10 \text{ A}$$

$$i_L(5 \text{ ms}) = I_x$$

$$I_x = 10 \left(1 - e^{-\frac{5}{4}}\right) = 7.135 \text{ A}$$

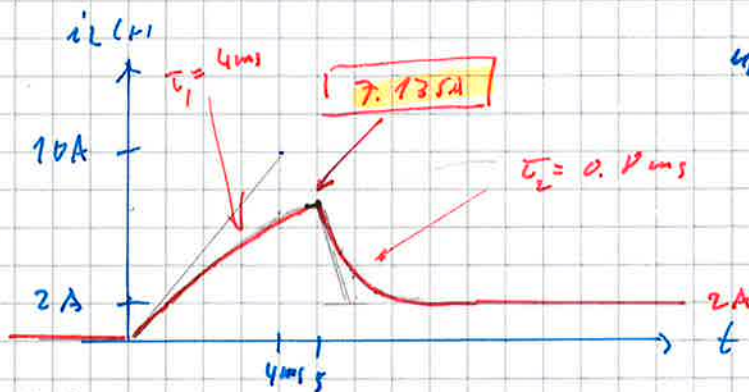


$$\tau_2 = 0.8 \text{ ms}$$

$$i_{L2}(5 \text{ ms}) = I_x$$

$$i_{L2}(\infty) = 2 \text{ A}$$

(2) $i_L(t)$



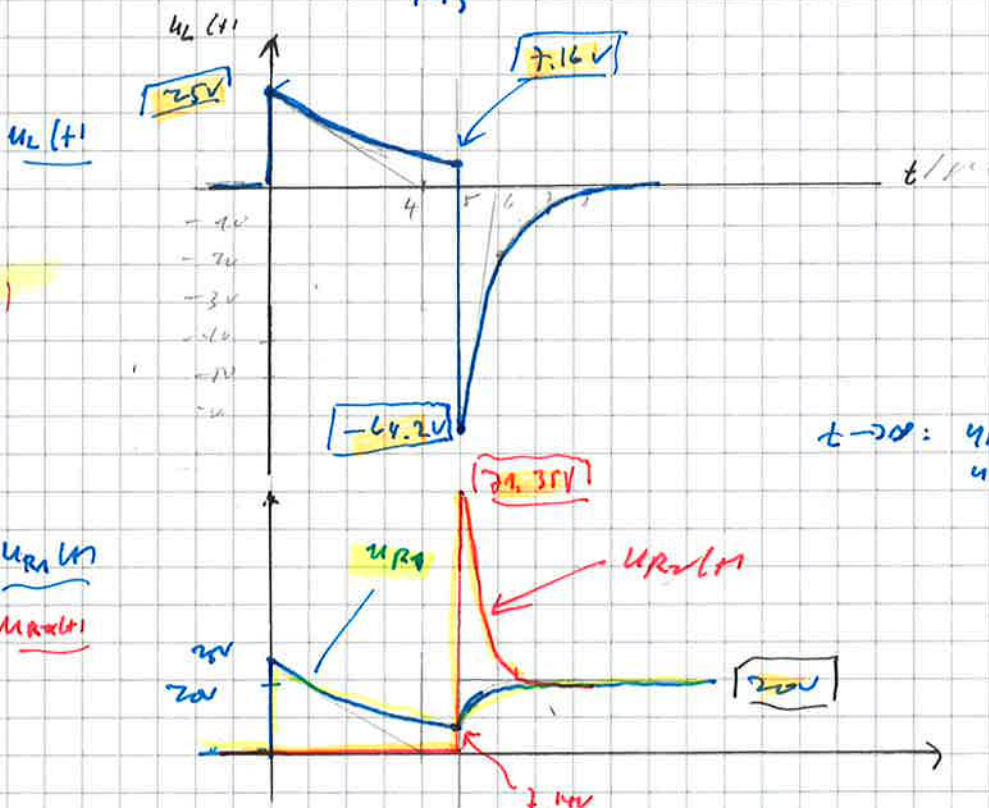
$$u_L = L \frac{di}{dt}$$

$$t=0: \frac{di}{dt} \Rightarrow u_L(0) = 25 \text{ V}$$

$$u_L(5 \text{ ms}): \frac{di}{dt} = \frac{7.135 - 2}{0.8 \text{ ms}}$$

$$\Rightarrow u_L = -64.2 \text{ V}$$

(2)

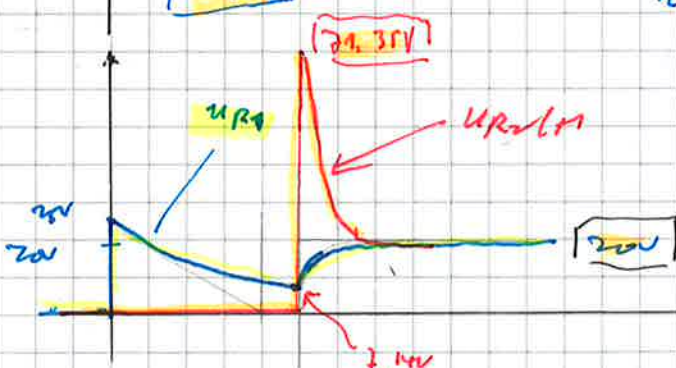


$$t \rightarrow \infty: u_{R1} = 20 \text{ V}$$

$$u_{R2} = 20 \text{ V}$$

(4)

$u_{R1}(t)$
 $u_{R2}(t)$

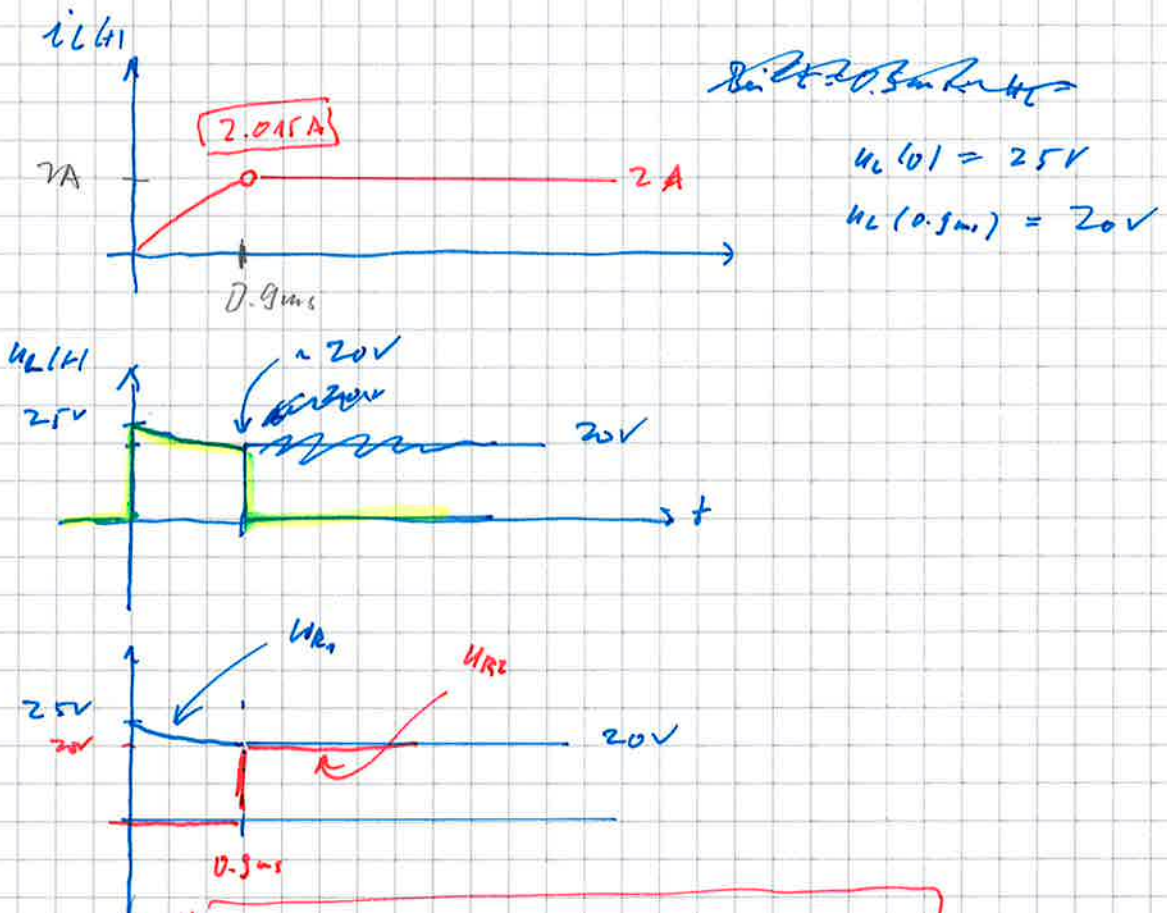


Bonus:

I_x berechnen: $0.9ms$

$$I_x = 10 \left[1 - e^{-\frac{0.9}{\tau}} \right] = 2.015 A$$

ist schon praktisch der Endwert



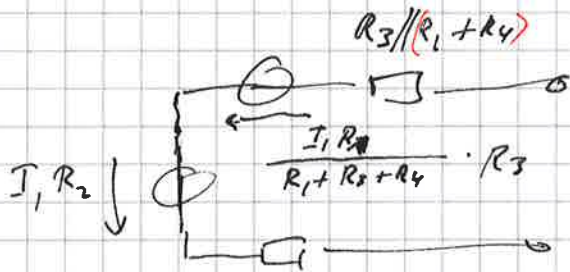
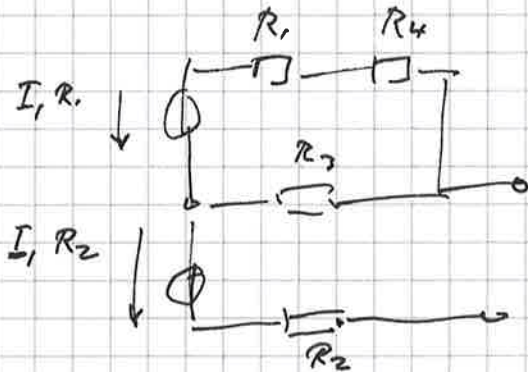
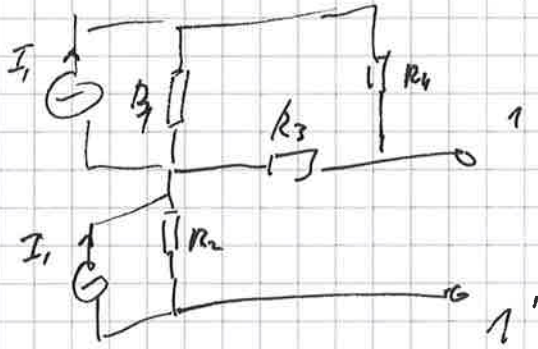
Ab $0.9ms$ ist Schaltung in Regel
 praktisch schon eingeschwenkt:

(dead beat Regel)

2.

(3)

$$R_{fe} = R_2 + R_3 // (R_1 + R_4) = 3.8 + 5 = 8.8 \Omega$$



$$U_{ge} = I_1 R_2 + \frac{I_1 R_1 \cdot R_3}{R_1 + R_3 + R_4} = I_1 \left[R_2 + \frac{R_1 R_3}{R_1 + R_3 + R_4} \right] = 23.75 \text{ V}$$

7

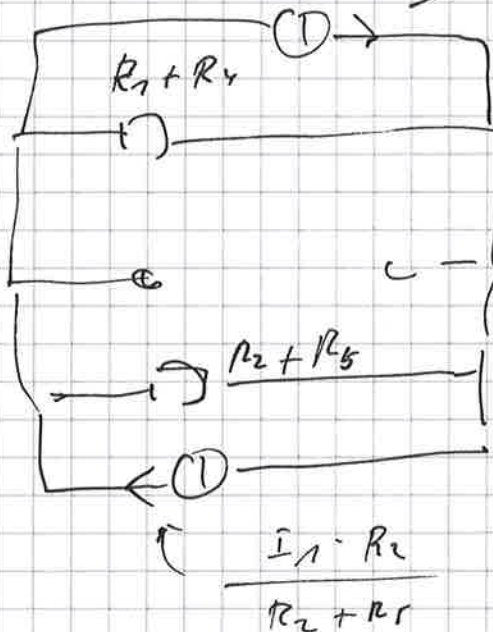
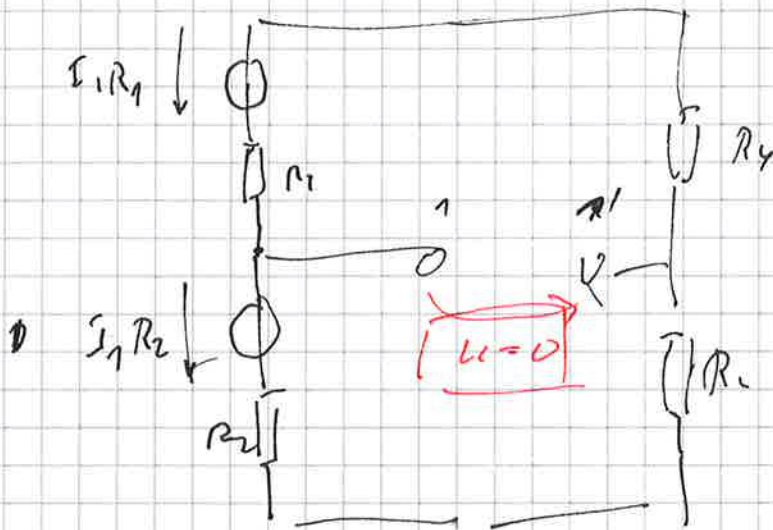
A

2.

b) z.B. $R_3 = 0$ (durch KS ersetzen) und
verknüpfung, dass Strom durch die KS Null ist.

Oder R_3 durch Induktivität ersetzen und
Verknüpfung, dass i_L Null ist.

Ersetzquelle man:



$$\frac{I_1 R_1}{R_1 + R_4}$$

$$I_{sc} = I_1 \left[\frac{R_1}{R_1 + R_4} - \frac{R_2}{R_2 + R_5} \right]$$

~~BA~~

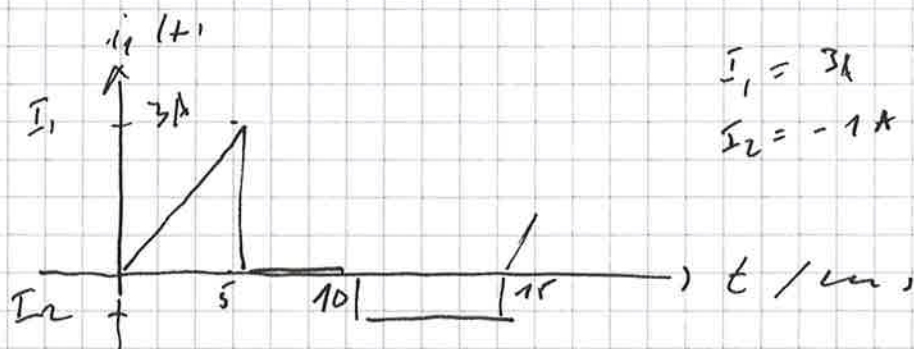
$$\frac{R_4}{R_1} + 1 = \frac{R_5}{R_2} + 1$$

$$\Rightarrow R_s = R_2 \cdot \frac{R_4}{R_1} = R_{sc}$$

✓

3.

15



$$I_1 = 3A$$

$$I_2 = -1A$$

$$\overline{i(t)} = \frac{5 \cdot I_1 / 2 + I_2 \cdot 5}{15} = \frac{I_1}{6} + \frac{I_2}{3} =$$

$$= \frac{I_1}{6} + \frac{I_2}{3} = 0.166A \quad (1)$$

$$\overline{i^2(t)} = \frac{I_1^2 \cdot 5}{3} + \frac{I_2^2 \cdot 5}{15} = \frac{I_1^2}{9} + \frac{I_2^2}{3}$$

$$\overline{i^2(t)} = \frac{I_1^2 \cdot 5}{3} + \frac{I_2^2 \cdot 5}{15} = \frac{I_1^2}{9} + \frac{I_2^2}{3}$$

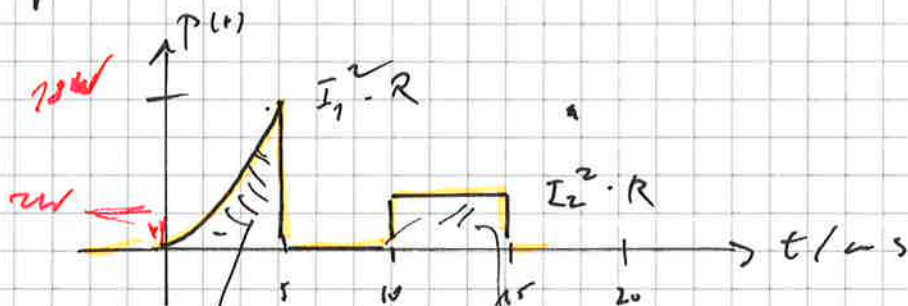
a)

$$= 1.33 A^2 \quad (2)$$

$$\Rightarrow \overline{i_{eff}} = 1.154 A$$

b)

$$p(t) = i^2 \cdot R$$

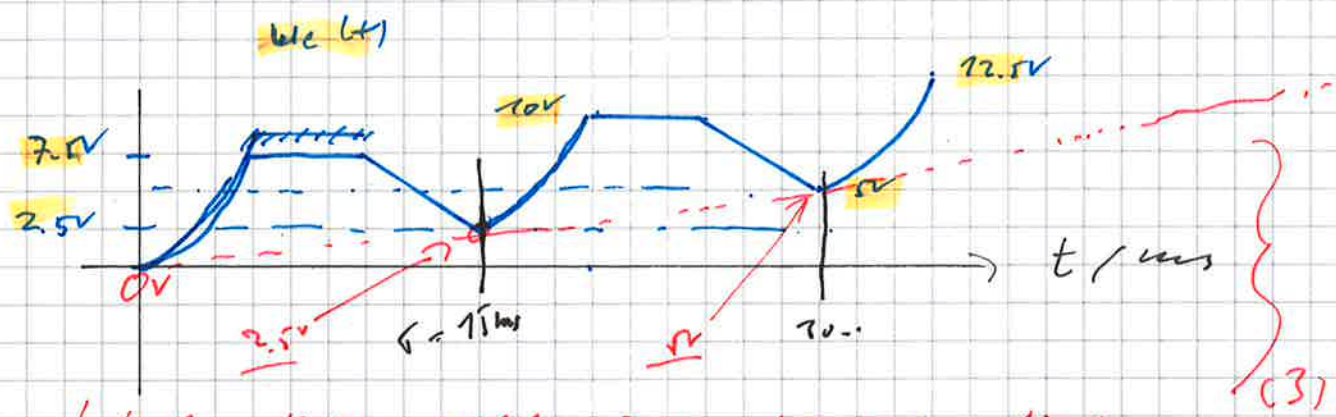
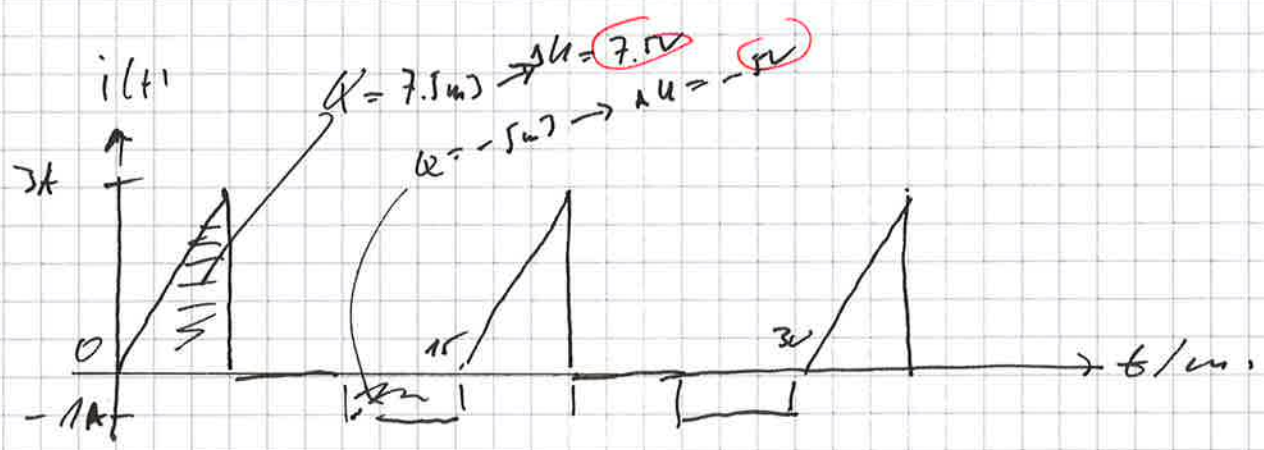


$$A_1 = I_1^2 \cdot R \cdot \frac{1}{3} \cdot 5ms, \quad A_2 = I_2^2 \cdot R \cdot 5ms$$

$$\overline{p(t)} = \frac{I_1^2 \cdot \frac{1}{3} \cdot 5ms + I_2^2 \cdot 5ms}{15ms} \cdot R = 2.666W \quad (3)$$

3.

c) $u_C(t) = \frac{1}{C} \int i(x) dx + U_0$



Wird 12V nicht DC-für i(t) läuft
 $u_C(t)$ down! (Integration)

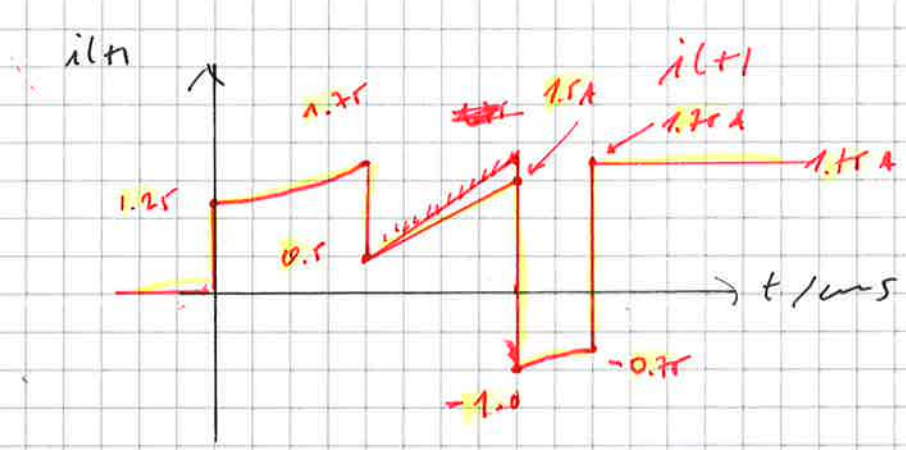
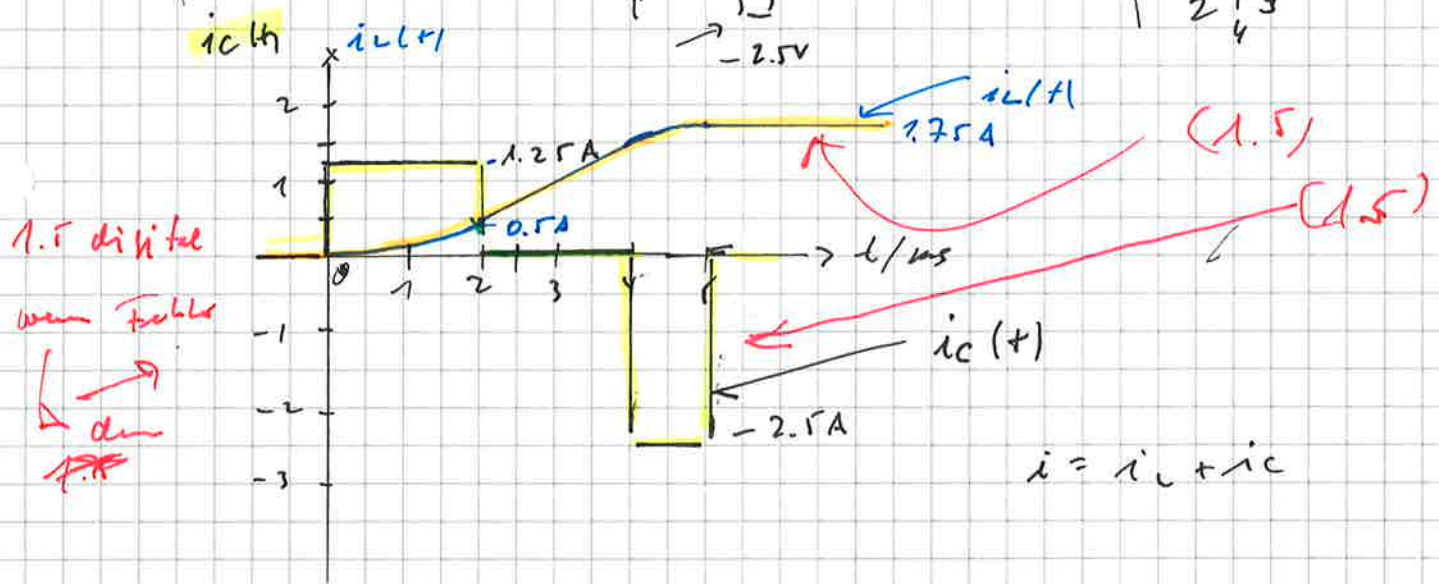
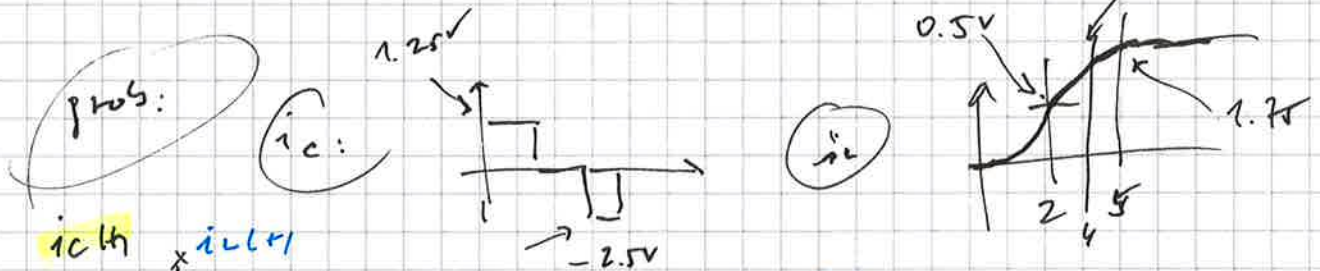
$i_{DC} = \overline{i(t)} = 0.16 \bar{C} \text{ A} \rightarrow \text{Offset pro Periode} \therefore \underline{\underline{+2.5V}}$

4.

Grundgesetze L/C:

$$u_L = L \cdot \frac{di_L}{dt} \rightarrow i_L(t) = \frac{1}{L} \cdot \int u_L(x) dx + I_{c0}$$

$$u_C = \frac{1}{C} \int i_C(x) dx \rightarrow i_C(t) = C \cdot \frac{du_C}{dt}$$



(4)
 $L = 10 \mu H$
 $C = 500 pF$

$$W = W_C + W_L = \frac{1}{2} C \cdot u_C^2 + \frac{1}{2} L \cdot i_L^2$$

t	i_L	u_C	W_{Ges}
2 μs	0.5 A	5 V	$1.25 + 6.25 = 7.5 \mu J$
4 μs	1.75 A	5 V	$11.25 + 6.25 = 17.5 \mu J$
6 μs	1.75 A	0 V	$15.3125 + 0 = 15.3125 \mu J$

(3)

57.

18

a) siehe Aufgabenblatt

$$\underline{\underline{I_0 = 0,125 \text{ A}}}$$

$$b) \underline{\underline{u_{eff} = 5 \text{ V}}}$$

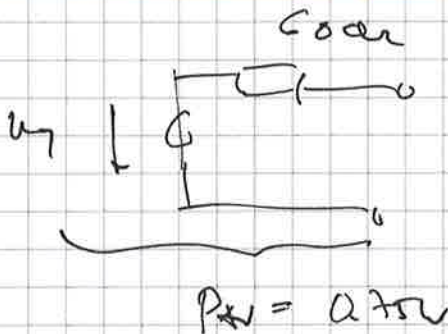
Bsp. Wert? Alle Frequenzen unendlich klein

also ...

$$\underline{\underline{u_{eff}^2 = 5^2 + \frac{3^2}{2} + \frac{4^2}{2} + \frac{5^2}{2} = 50}}$$

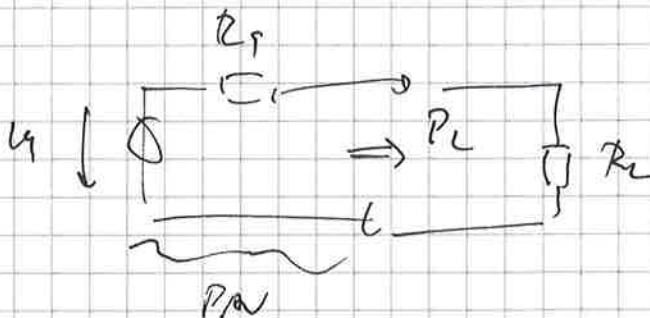
$$\underline{\underline{u_{eff} = \sqrt{50} = 7,07 \text{ V}}}$$

c)



$$P_N = \frac{u_g^2}{4R_L}$$

$$\Rightarrow \underline{\underline{u_g = \sqrt{4R_L \cdot P_N} = 42,43 \text{ V}}}$$



$$P_L < P_N$$

$$v = \frac{P_L}{P_N} = 0,64$$

→ 2 Lösungen

gesuch. H.
(oder graphisch, siehe
Rechenblatt)

$$x_{1,2} = \frac{2}{v} - 1 \pm \sqrt{\left(\frac{2}{v} - 1\right)^2 - 1}$$

$$v = \frac{4}{2+x + \frac{1}{x}} = \frac{P_L}{P_N}$$

$$\left. \begin{array}{l} x_1 = 4 \\ x_2 = 0,25 \end{array} \right\}$$

$$R_{L1} = 240 \Omega$$

$$R_{L2} = 150 \Omega$$

- d) R_5 entfernen und Ersatzquellen links + rechts
 ... (siehe Skizze) : gibt "direkt":

$$\underline{I_r} = \frac{U_{q1} - U_{q2}}{R_{q1} + R_5} = \frac{U_1 \left[\frac{R_2}{R_1 + R_2} - \frac{R_4}{R_3 + R_4} \right]}{R_1 // R_2 + R_3 // R_4 + R_5}$$

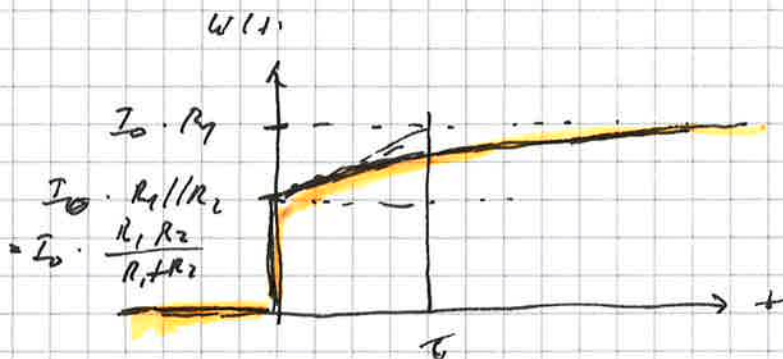
- e) Für Spry ist C Kurzschluss, d.h. $u_C = 0$

→ Stromkling auf R_1 und R_2

$$i_{R2}(0) = I_0 \cdot \frac{R_1}{R_1 + R_2}$$

$t \rightarrow \infty$: C vollgelad, $i_C = 0$, nur noch R_1

$$\rightarrow \text{D } u(\infty) = I_0 \cdot R_1$$



$$\underline{\tau = C \cdot (R_1 + R_2)}$$